

Fraud & Anomaly Detection Implementation Plan

Complete architectural and algorithmic blueprint for detecting, triaging, and mitigating marketplace fraud across 20 distinct risk vectors spanning Users/Hosts, Transactions, and Listings.

1. Executive Summary & Architecture Overview

SpaceLoop is a peer-to-peer physical micro-space marketplace operating under the core lifecycle: **Host** → **Listing** → **Search/Match** → **Booking** → **Access** → **Checkout/Escrow**. To protect marketplace liquidity, prevent financial loss, stop synthetic reputational manipulation, and safeguard physical premise security, SpaceLoop operates a multi-layered, deterministic Trust & Safety engine augmented by multi-tier explainable AI forensics. This document details the complete end-to-end implementation for all **20 audit checklist rules** divided across three pillars.

Layer 1: Deterministic Rule Gates	Layer 2: Network Graph & Stats	Layer 3: Multi-Tier Forensic AI
Fast-path O(1) checks executed inline during user signup, listing publish, and checkout. Blocks critical threats deterministically with zero latency.	Graph collusion analysis, modified Z-score statistical price bounds, device fingerprint clustering, and Haversine geo-velocity checks.	Groq LLaMA 3.3 70B → Gemini 2.5 Flash → Deterministic rule template synthesis. Generates natural language factual briefs for human moderators.

2. Canonical 20-Rule Detection Matrix

Standardized taxonomy of all 20 audit rules implemented across the SpaceLoop Trust & Safety subsystem:

#	Pillar / Vector	Signal Code	Detection Methodology	Severity	Action Gate
1	User/Host: Suspicious Creation	SUSPICIOUS_ACCOUNT_CREATION	Velocity burst (>3 accounts in 1h from same IP/subnet) or clustered synthetic profiles.	HIGH	Hold / Verify
2	User/Host: Disposable Email	DISPOSABLE_INVALID_EMAIL	Email matches disposable/burner domains (Mailinator, TempMail, GuerrillaMail, etc.).	HIGH	Require Work/SSO
3	User/Host: Shared Identifiers	SHARED_IDENTIFIERS_MULTI_ACCOUNT	Multiple accounts sharing phone, Aadhaar hash, UPI VPA, or device fingerprint.	CRITICAL	Restrict / Block
4	User/Host: Abnormal Login	ABNORMAL_LOGIN_DEVICE	Novel hardware fingerprint, sudden user-agent shifts, or impossible travel velocity.	MEDIUM	Enforce MFA
5	User/Host: Sudden Profile Change	SUDDEN_PROFILE_MUTATION	Modifying payout UPI/bank or email <24h before high-value booking or escrow payout.	HIGH	Hold Escrow 24h
6	User/Host: Fake/Duplicated Listings	FAKE_DUPLICATED_LISTINGS	Single host publishing clone listings across distinct addresses to game search.	HIGH	Unpublish / Flag
7	User/Host: Suspicious Host Behavior	SUSPICIOUS_HOST_BEHAVIOR	Unverified host (no Discom/Aadhaar) posting luxury rates or >70% rejection rate.	MEDIUM	Request KYC
8	Transaction: High Booking Freq.	HIGH_BOOKING_FREQUENCY	Excessive reservation velocity (>3 in 15m, >8 in 24h) by a single seeker.	HIGH	Rate Limit / Hold
9	Transaction: Failed Payments Burst	REPEATED_FAILED_PAYMENTS	>= 3 payment gateway/UPI failure attempts in 15m indicating credential/card testing.	HIGH	Cooldown 30m
10	Transaction: Unusual Amount	UNUSUAL_BOOKING_AMOUNT	Total booking price >3.5x space historical median or >Rs 50,000 without KYC.	HIGH	Manual Review
11	Transaction: Rapid Churn Cycles	RAPID_BOOKING_CANCELLATION	Reservations created and cancelled <5m, or >50% cancellation velocity (slot blocking).	HIGH	Temporary Lockout
12	Transaction: Shared Payment ID	SHARED_PAYMENT_IDENTITY	Distinct seeker accounts paying with or withdrawing to the exact same UPI VPA.	CRITICAL	Hold Escrow / Ban
13	Transaction: Abnormal Geo/Device	ABNORMAL_GEO_DEVICE_CHANGE	Booking IP distance >1,000km from physical space geofence without travel window.	MEDIUM	Verification Gate

#	Pillar / Vector	Signal Code	Detection Methodology	Severity	Action Gate
14	Transaction: Suspicious Refunds	SUSPICIOUS_REFUND_PATTERNS	Recurring micro-escrow disputes or false condition damage claims across spaces.	HIGH	Flag for Mediation
15	Listing: Duplicate Description	DUPLICATE_LISTING_DESCRIPTION	Listing description has >= 85% textual token/cosine similarity to existing space.	HIGH	Reject Duplication
16	Listing: Similar Images/Content	SIMILAR_IMAGES_CONTENT	Re-use of identical photo assets or cloned amenity fingerprint across listings.	HIGH	Flag for Audit
17	Listing: Unrealistic Pricing	UNREALISTIC_PRICING	Hourly rate deviates by >3.0 Modified Z-Score / IQR from neighborhood benchmark.	HIGH	Price Calibration
18	Listing: Conflicting Info	CONFLICTING_LISTING_INFO	Contradictory physical attributes (e.g. sqft < 50 with capacity > 20, desk with 50 ppl).	MEDIUM	Block Publishing
19	Listing: Repeated Across Accounts	REPEATED_LISTINGS_ACROSS_ACCOUNTS	Identical physical address, GPS coords within 15m, or same Discom CA meter across hosts.	CRITICAL	Discom Deed Verify
20	Listing: Unusual Host/Activity	UNUSUAL_HOST_LISTING_ACTIVITY	Dormant space (>60 days inactive) sudden reactivation with price spikes or changed bank.	MEDIUM	Re-verify Ownership

3. Pillar Deep Dive & Forensic Algorithms

Detailed technical specification of algorithmic gates and deterministic detection engines:

Pillar 1: User & Host Integrity

3.1 Pillar 1: Users & Hosts Intelligence

- **Disposable Email Detection:** Employs an exact tokenized domain parser and blocklist of 150+ burner providers (e.g., mailinator, guerrillamail, tempmail, 10minutemail, throwaway, sharklasers). High-risk accounts are quarantined from listing spaces.
- **Multi-Account Identifier Matching:** Checks phone numbers, SHA-256 tokenized Aadhaar identity hashes, Discom CA utility numbers, and client hardware fingerprints. When collision cardinality ≥ 2 , triggers CRITICAL identity deception alerts.
- **Sudden Profile Mutation Defense:** When a host mutates their UPI VPA or bank payout account, a 24-hour escrow release hold is deterministically enforced to prevent account takeover evacuation.

Pillar 2: Transaction & Micro-Escrow Protections

3.2 Pillar 2: Transaction & Payment Abuse Mitigation

- **High Booking Velocity & Slot Blocking:** Rolling 15-minute sliding window checks for burst reservations. If a seeker creates > 3 bookings or $> 50\%$ cancellation velocity in 7 days, temporary reservation cooldowns are applied.
- **Repeated Failed Payments (Card/UPI Testing):** Tracks failed checkout transactions. ≥ 3 failures within 15 minutes triggers a 30-minute automated rate-limit lockout to neutralize automated payment probing.
- **Shared Payment Identity:** Aggregates payee UPI VPAs across accounts. Detects collusion rings where multiple buyer personas channel micro-escrow or bookings into a single financial nexus.
- **Abnormal Location Shifts:** Evaluates Haversine distance between client IP geocoordinates and physical space geofences. Discrepancies $> 1,000$ km without prior itinerary history require interactive TOTP/OTP verification.

Pillar 3: Listing Veracity & Premise Accuracy

3.3 Pillar 3: Listing & Physical Premise Verification

- **Content & Description Duplication:** Implements tokenized Jaccard similarity and character-level Levenshtein distance. Matches $\geq 85\%$ similarity against any active space in the catalog are immediately flagged.
- **Statistical Outlier Pricing:** Calculates Modified Z-Score using Median Absolute Deviation (MAD): $M_i = 0.6745 * (x_i - Median) / MAD$. Modified Z-Scores > 3.0 trigger automated price abnormality reviews.
- **Physical Attribute Consistency:** Evaluates capacity density: $sqft / max_capacity \geq 12 sqft/person$. Impossible combinations (e.g. 50 sqft with 20 occupants, or storage units claiming desks) fail structural validation.
- **Cross-Account Property Re-listing:** Cross-references Discom electricity meter numbers and GPS coordinates (within 15m radius). Prevents bad actors from spinning up fresh host accounts to evade past property bans.

4. Mathematical Risk Scoring & Triage Engine

Diminishing Marginal Risk Formulation:

To avoid linear clipping and excessive false positives from multiple low-severity signals, the composite risk score $R \in [0.0, 1.0]$ is calculated using non-linear independent probability accumulation:

$$R = 1.0 - \prod (1.0 - w_i)$$

where w_i is the normalized weight of triggered signal i .

Statistical confidence is scaled monotonically with signal cardinality: $C = \min(0.98, 0.65 + 0.10 * N)$.

Risk Tier	Score Range	Automated Action	Human Moderator SLA
NORMAL	0.00 – 0.24	Allow immediately. Telemetry logged.	No review needed
UNUSUAL	0.25 – 0.49	Request step-up verification (OTP/TOTP/DigiLocker).	Queue for spot-check (24h)
SUSPICIOUS	0.50 – 0.74	Hold transaction / Escrow lock. Alert operations.	Priority triage (<2h)
HIGH RISK	0.75 – 1.00	Restrict action / Unpublish listing / Quota lockout.	Immediate intervention (<15m)

5. End-to-End Implementation Deliverables

Targeted modifications across backend services, API contracts, and React frontend:

File / Component	Change Scope	Key Functional Enhancements
backend/.../signals.py	Signal Taxonomy Registry	Registers all 20 canonical signals with explicit severities, base weights, and category tags.
backend/.../rules.py	Deterministic Rule Gates	Adds <code>evaluate_user_host_rules()</code> (7 checks), enhances <code>evaluate_booking_rules()</code> (7 checks), and <code>evaluate_listing_rules()</code> (6 checks).
backend/.../engine.py	Central Orchestration Engine	Implements <code>evaluate_user_host()</code> , batch auditing engine, graph cycle integration, and multi-tier LLM forensic narrative generation.
backend/.../trust_safety.py	REST API Blueprint	Adds <code>POST /api/v1/trust-safety/scan-all</code> for on-demand audits, category filters for triage queries, and manual override actions.
frontend/src/types.ts	TypeScript Interfaces	Defines <code>RiskCategory</code> , <code>SignalDefinition</code> , and extended <code>RiskAssessment</code> models.
frontend/.../TrustSafetyPage.tsx	Operations Dashboard	Tabbed navigation for Users/Hosts, Transactions, and Listings; 1-click audit button; live signal pills and evidence drawers.
tests/test_trust_safety_rules_20.py	Comprehensive Test Suite	Unit and integration tests covering all 20 rules individually and end-to-end.

6. Verification & Quality Assurance Protocol

Rigorous multi-stage verification to ensure zero regressions across SpaceLoop's production flow:

Quality Assurance & Deployment Checklist
Phase 1: Automated Unit & Rule Verification

- Execute `tests/test_trust_safety_rules_20.py` to test all 20 rules with synthetic attack vectors.
- Execute full regression suite `tests/test_trust_safety.py` and `test_all_features_functional.py` (14/14 audits).

Phase 2: API Contract & Batch Scan Testing

- Verify `POST /api/v1/trust-safety/scan-all` generates assessments across all active spaces, bookings, and users.
- Verify manual moderation overrides (`override_allow`, `confirm_block`, `hold_escrow`, `request_id`) transition state correctly.

Phase 3: Frontend Production Build & Deployment

- Execute `npm run build` in `frontend/` to verify TypeScript types and Tailwind compilation.
- Synchronize generated build artifacts to `public/` for Render production static file serving.

Architectural Approval SpaceLoop Core Engineering Team	Compliance Verification India Stack & DPDP Act 2023	Target Execution Date March 2026 // Production Ready
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